

Kanishk Bhati

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SUMMARY

Backend-focused full-stack engineer building constraint-based scheduling services, authentication and session layers, and query-level performance work in Spring Boot, PostgreSQL, and Redis — including REST API design, Dockerized deployment, and monitoring with Prometheus and Grafana.

TECHNICAL SKILLS

Languages	Java, TypeScript, JavaScript, Python
Backend	Spring Boot, NestJS, Express.js, REST API design, JPA/Hibernate, JWT authentication, session management, WebSockets, RBAC
Optimization	Timefold Solver, constraint modeling, vehicle routing (VRP), greedy heuristics
Data & Cache	PostgreSQL, MySQL, SQLite, MongoDB, Redis, query and schema optimization
Frontend	Angular (Nx), React
DevOps	Docker, Docker Compose, multi-stage builds, GitHub Actions, Maven, Git, AWS (S3, SES), Prometheus, Grafana, Resilience4j

PROFESSIONAL EXPERIENCE

28One **Aug 2024 – Present**
Software Development Engineer (Full Stack) *Gurugram, Haryana*

- Designed and built an Auto-Planner service (vehicle routing / inspection scheduling) in Spring Boot, moving inspector allocation from a greedy nearest-resource heuristic to constraint-based optimization with Timefold Solver.
- Modeled domain constraints including travel time, inspection duration, capacity limits, and unassigned-booking penalties, and configured solver termination strategies (time-based and unimproved-spent limits).
- Optimized database schema and query performance, reducing data-retrieval latency by 35%.
- Exposed REST endpoints for dynamic planning and re-planning of approved bookings, and containerized the service with multi-stage Docker builds for environment-based deployments.

Coding Ninjas **Feb 2024 – Jul 2024**
Software Development Engineer Intern (Full Stack) *Gurugram, Haryana*

- Optimized website performance, reducing load time by 30% using caching and lazy loading.
- Automated leaderboard updates using cron jobs and event-driven triggers, increasing leaderboard engagement by 60%.
- Revamped the course navigation flow, contributing to a 15% increase in user retention.

PROJECTS

MarketMate — Distributed Marketplace Platform (personal project) [Live](#) · [Backend](#) · [Frontend](#) · [Chat](#) · [Infra](#)

- Built and deployed a marketplace as separate services — Spring Boot REST API, Angular (Nx) SSR frontend, admin app, and a NestJS chat service — across custom subdomains with SSL, CORS/CSP, and CDN-backed delivery.
- Implemented authentication with JWT in httpOnly cookies, PostgreSQL-backed sessions, and a Redis validation cache, so two independently deployed services could trust the same session without routing every authentication check through the main API.
- Integrated AWS S3 for media storage and AWS SES (SDK v2) for OTP, verification, and notification email with domain-level DKIM, guarded by Resilience4j retry and circuit breaker policies.
- Containerized the stack with Docker Compose and instrumented it with Prometheus and Grafana.

Alphabet Music API [API](#) · [Android App](#)

- Built a music management REST API and Android client using Node.js, Express.js, MongoDB, and Java, with JWT-based authentication and Multer multipart file handling for uploads.

WRITING

Engineering notes at kkbhati07.github.io — on JWT + Redis session validation, and on choosing between retry, circuit breaker, and fallback when a dependency fails.

EDUCATION

IIMT Engineering College, Meerut, U.P. — B.Sc. Computer Science **Aug 2016 – Jul 2019**
Coding Ninjas, Gurugram, Haryana — Web Development (certification program) **Aug 2022 – Dec 2023**